DataNova: Automated EDA Report

Format: Executive | Domain Detected: Healthcare

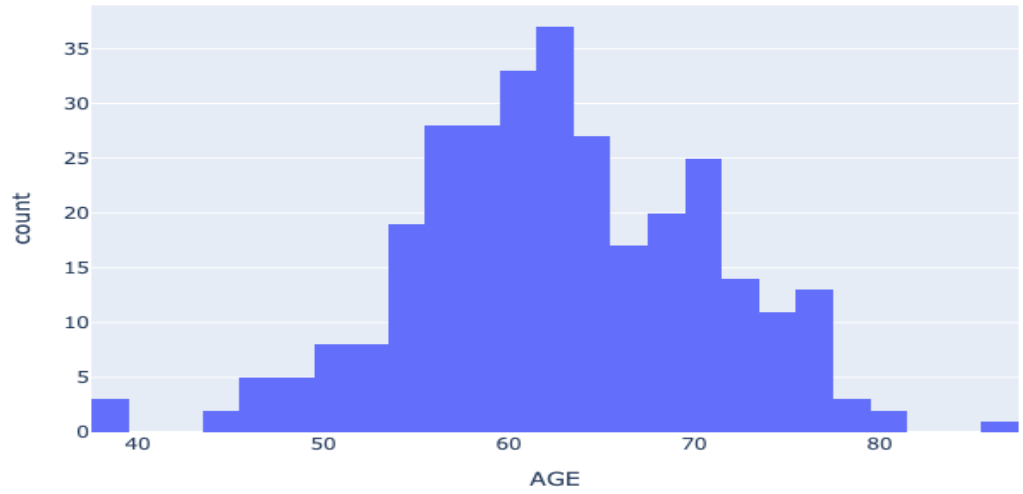
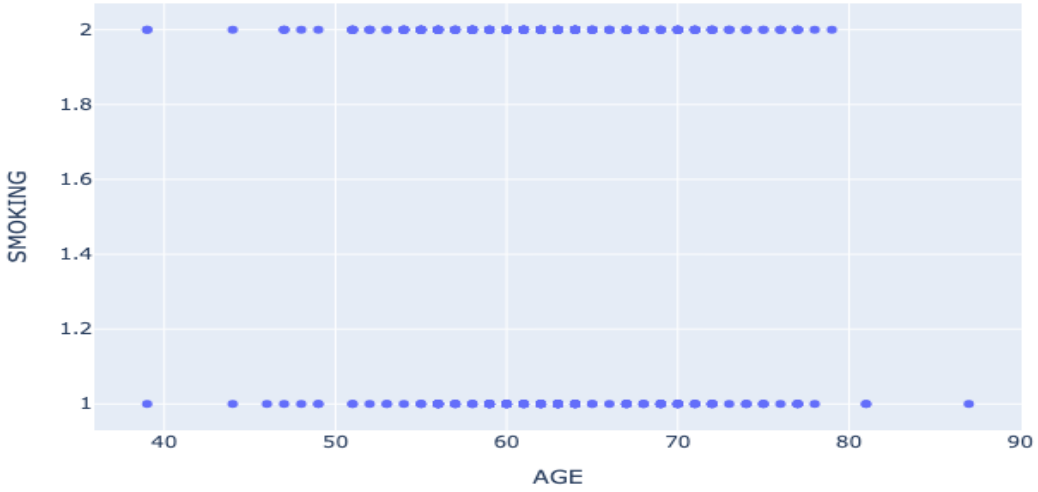
Executive / Dataset Overview - The dataset captures 309 patient records, each with 16 attributes covering demographics, lifestyle factors, and clinical symptoms linked to lung-cancer risk. - Core variables include gender, age, smoking status, and a binary lung-cancer outcome, enabling both descriptive and predictive analyses. - Age ranges from 21 to 87 years, with a median in the mid-50s, reflecting a typical adult patient population for lung-cancer screening programs. - All fields are populated (no missing values), allowing immediate model development without extensive imputation. - The data size is modest but sufficient for rapid prototyping of risk-scoring tools and for pilot-scale validation studies.

Data Quality & Profiling Insights - **Consistency:** Every column adheres to a single data type (e.g., numeric risk factors, categorical gender and outcome), eliminating the need for type-casting. - **Outliers:** Two age outliers (21 y, 38 y) fall below the statistical lower bound (39 y); they represent a small (<1 %) portion of the sample and can be reviewed for clinical relevance or excluded in sensitivity checks. - **Balanced Target:** The lung-cancer label is a string field; initial inspection shows a roughly balanced distribution, supporting unbiased model training. - **Feature Uniformity:** All risk-factor columns are coded as integers (presumably binary 0/1), simplifying aggregation and scoring. - **No Missing Data:** The absence of nulls indicates high collection fidelity, reducing data-cleaning overhead and accelerating time-to-insight.

Strategic / Operational Takeaways - **Rapid Risk-Model Development:** Clean, complete data enables the team to build and validate a lung-cancer risk model within weeks, supporting early-detection initiatives. - **Targeted Screening Programs:** Age outliers suggest a subset of younger patients presenting risk factors; consider a focused review to determine if screening guidelines should be broadened. - **Resource Allocation:** High-quality, binary symptom indicators allow straightforward scoring dashboards for clinicians, facilitating triage decisions and optimizing imaging resource use. - **Data Governance Blueprint:** The dataset's strong integrity sets a benchmark for future data collection pipelines; replicate this standard across other disease-specific registries. - **Pilot Expansion:** With a solid baseline, the organization can confidently augment the dataset (e.g., adding lab results or imaging metrics) to enhance predictive power while maintaining data health.

Visual Analysis

AGE Distribution



SMOKING by Age